VIT
Vellore Institute of Technology
(Deemed to be University under section 3 of UGC Act, 1956)
CHENNAI

Reg. Number:	1605

Continuous Assessment Test (CAT) – II OCTOBER 2025

Programme	:	B.Tech. (CSE) with all specializations.	Semester	:	Fall Semester 2025-2026
Course Code & Course Title	:	BCSE204L Design and Analysis of Algorithms	Class Number	:	CH2025260101320,CH2025260101448 CH2025260101449,CH2025260101593 CH2025260101595,CH2025260101598 CH2025260102084,CH2025260102085 CH2025260102086,CH2025260102087 CH2025260102088
Faculty	:	Dr Aarthi B, Dr Jannath Nisha O S, Dr Ranjith Kumar M, Dr Raja M, Dr Indira B, Dr Lekshmi K, Dr Sivaranjani, Dr Revathi A R, Dr Helen Vijitha P, Dr Sarita Kumari, Dr kaja Mohideen A	Slot	:	G2+TG2
Duration	:	90 Minutes	Max. Mark		50

General Instructions:

- Write only your registration number on the question paper in the box provided and do not write other information
- If any assumptions are required, assume the same and mention those assumptions in the answer script.
- Your answer to all the questions should have both the design and the analysis components.
- The design component should consist of logic to develop and illustrate the pseudocode.
- The analysis component should consist of the computation of T(n) and time complexity.

Answer all questions

Q. No	Description	Marks	CO	BT Level
1	A network security company is designing an efficient encoding scheme for transmitting encrypted packets that contain different types of elements: Alphabets (P–T), numeric symbols (1, 2, 3), and protocol signals (α, β). The observed frequencies of occurrences in a packet dataset are as follows (Element – Frequency): P – 40, Q – 18, R – 10, S – 12, T – 15, 1 – 30, 2 – 22, 3 – 8, α – 25, β – 20 a. Construct the Huffman tree for the given set of elements. (2 marks) b. Generate the Huffman codes for each element. (2 marks) c. Calculate the average number of bits per symbol in this coding scheme. (2 marks) d. Compare this result with fixed-length encoding. (2 marks) e. Analyze which scheme is more efficient for transmitting encrypted packets and justify. (2 marks)	10	CO2	K1
2	You are given an N × N checkerboard and a single checker. The checker starts at any square in the bottom row (row N) and must reach some square in the top row (row 1). At each step, the checker may move to one of the following squares (if valid): • The square immediately above. • The square one up and one to the left (unless already in the leftmost column). • The square one up and one to the right (unless already in the rightmost column). Each move from square x to square y yields p(x, y) rupees, which may be positive or negative. The values p(x, y) are known for all legal moves. The objective is to determine the maximum amount of money the checker can collect while moving from the bottom row to the top row.	10	CO2	K2

	a. Apply the backtracking strategy to find the optimal solution. (Illustration: 4 marks, Pseudocode: 4 marks) b. Analyze the time complexity of your pseudocode. (2 marks)			
3	In data compression, a circular buffer is often used to store characters in a repeating sequence, where the logical start of the buffer can be any position in the string. For a circular string T of length m, there are m possible rotations that represent the same buffer. To ensure consistency in compression and indexing, it is useful to select a single canonical representation, the lexicographically smallest rotation. a. Design a pseudocode using rolling-hash technique to identify the lexicographically smallest rotation of T in expected time of $O(m \log m)$. (6 marks) b. Prove that your pseudocode correctly determines the canonical rotation. (5 marks) c. Derive the time complexity of your pseudocode. (4 marks)	15	CO3	K3
4	Six students A, B, C, D, E, F are being evaluated over a series of internal tests. Each student's average mark (lower – upper) falls in one of the following ranges: Student | A | B | C | D | E | F Range of average marks | 80 – 90 | 70 – 85 | 60 – 78 | 75 – 88 | 50 – 72 | 65 – 82 An influence graph G = (V, E) is to be built with V = {A, B, C, D, E, F}. Each directed edge (u → v) represents "u influences v". The base weight for u → v is calculated as follows: w(u, v) = (lower bound of v's range) – (upper bound of u's range) The negative weight in a graph indicates that the predicted mark can drop (risk factor) while a positive weight indicates a possible gain. The faculty identifies the following potential influences (edges) : A → B, A → C, B → D, B → E, C → E, C → F, D → A, D → F, E → B, F → C a. Compute the initial weight of each edge using the above formula. (3 marks) b. Design an appropriate graph algorithm to detect the presence of any negative weight cycle. Illustrate the distance updates at each iteration of the algorithm. If a negative weight cycle is found, use predecessor tracing to list the vertices of one such cycle in the correct order. (Pseudocode: 4 marks, Illustration: 4 marks) c. Derive the time complexity of your algorithm. (4 marks)	15	CO3	K3

**********All the best **********